

6.1. (a) 输出方程:

$$\begin{aligned} Z &= \overline{Q\bar{X} + \bar{Q}X} \\ &= QX + \overline{QX} \end{aligned}$$

激励方程: $J=X, K=\bar{X}$

状态方程: $Q^{n+1} = X\bar{Q}^n + \bar{X}Q^n = X$

(b)

Q^n	X	Q^{n+1}	Z
0	0	0	1
0	1	1	0
1	0	0	0
1	1	1	1



6.2. (a) 激励方程:

$$D_1 = \overline{Q_1 Q_2 Q_3}$$

$$D_2 = Q_1$$

$$D_3 = Q_2$$

状态方程: $Q_1^{n+1} = \overline{Q_1^n Q_2^n Q_3^n}$

$$Q_2^{n+1} = Q_1^n$$

$$Q_3^{n+1} = Q_2^n$$

(b)	Q_1^n	Q_2^n	Q_3^n	Q_1^{n+1}	Q_2^{n+1}	Q_3^{n+1}
	0	0	0	1	0	0
	0	0	1	1	0	0
	0	1	0	1	0	1
	0	1	1	1	0	1
	1	0	0	1	1	0
	1	0	1	0	1	0
	1	1	0	0	1	1
	1	1	1	0	1	1

6.3 (a) 激励方程:

$$J_1 = \overline{Q_2^n Q_3^n}, k_1 = 1$$

$$J_2 = Q_1^n, k_2 = \overline{Q_1^n Q_3^n} = Q_1^n + \overline{Q_3^n}$$

$$J_3 = Q_1^n Q_2^n, k_3 = Q_1^n$$

状态方程:

~~$$Q^{n+1} = J\overline{Q^n} + KQ^n$$~~

$$Q_1^{n+1} = \overline{Q_1^n Q_2^n Q_3^n}$$

$$Q_2^{n+1} = Q_1^n \overline{Q_2^n} + (\overline{Q_1^n} Q_3^n) Q_2^n$$

$$Q_3^{n+1} = Q_1^n Q_2^n \overline{Q_3^n} + \overline{Q_1^n} Q_3^n$$

(b)

Q_1^n	Q_2^n	Q_3^n	Q_1^{n+1}	Q_2^{n+1}	Q_3^{n+1}
0	0	0	1	0	0
0	0	1	1	0	0
0	1	0	1	0	0
0	1	1	1	0	0
1	0	0	0	1	0
1	0	1	0	0	0
1	1	0	0	1	1
1	1	1	0	0	0

6.4(a) 激励方程: $J_1=1, k_1=1$

$$J_2=Q_1, k_2=1$$

$$J_3=Q_1Q_2, k_3=1$$

状态方程: $Q_1^{n+1} = \overline{Q_1^n}$

$$Q_2^{n+1} = Q_1^n \overline{Q_2^n}$$

$$Q_3^{n+1} = Q_1^n Q_2^n \overline{Q_3^n}$$

(b)	Q_1^n	Q_2^n	Q_3^n	Q_1^{n+1}	Q_2^{n+1}	Q_3^{n+1}
	0	0	0	1	0	0
	0	0	1	1	0	0
	0	1	0	1	0	0
	0	1	1	1	0	0
	1	0	0	0	1	0
	1	0	1	0	1	0
	1	1	0	0	0	1
	1	1	1	0	0	0